Methods and Results (draft)

Working draft assembled from the analyses completed in this repository. Tables are generated from the results CSVs and traceable to their source files; figures are the rendered outputs in `reports/`. Prose is descriptive of what was done and what the numbers show. Sentences marked `[Interpretation:]` are placeholders where the author's interpretation belongs and are not asserted here. Literature and citations are out of scope for this draft.

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2. Methods

2.1 Study area and interpreted reference

The study area is the western Great Lakes region spanning Wisconsin, Minnesota, and Michigan. The reference data are interpreted land-cover and change cells produced with the CKIT-RF interpreter workflow: for each cell an analyst draws training polygons over high-resolution imagery, a per-cell Random Forest is trained on those labels, and the result is a wall-to-wall classified raster on the cell footprint (EPSG:5070, 10 m). The interpreted set used here is 180 cells (36 cells in each of five NAIP acquisition brackets), of which 72 were independently interpreted by more than one reviewer.

Where a cell was interpreted by more than one reviewer, a single interpretation per cell is selected by adjudication, recorded in `exports/truth_selections.csv`, so each cell contributes exactly one reference and none is double counted. The 72 multi-interpreted cells are also retained separately for the inter-interpreter reliability analysis (Section 2.7), where each is scored as a reviewer pair.

The reference cells fall within the watersheds of seven park units of the National Park Service Great Lakes Inventory and Monitoring Network (GLKN): Apostle Islands, Grand Portage, Isle Royale, Mississippi National River and Recreation Area, Saint Croix, Sleeping Bear Dunes, and Voyageurs. Two of the nine GLKN units, Indiana Dunes and Pictured Rocks, fall outside the study grid extent and are not used. Figure 1 maps the study region, the grid extent, the park units, and the interpreted cells.

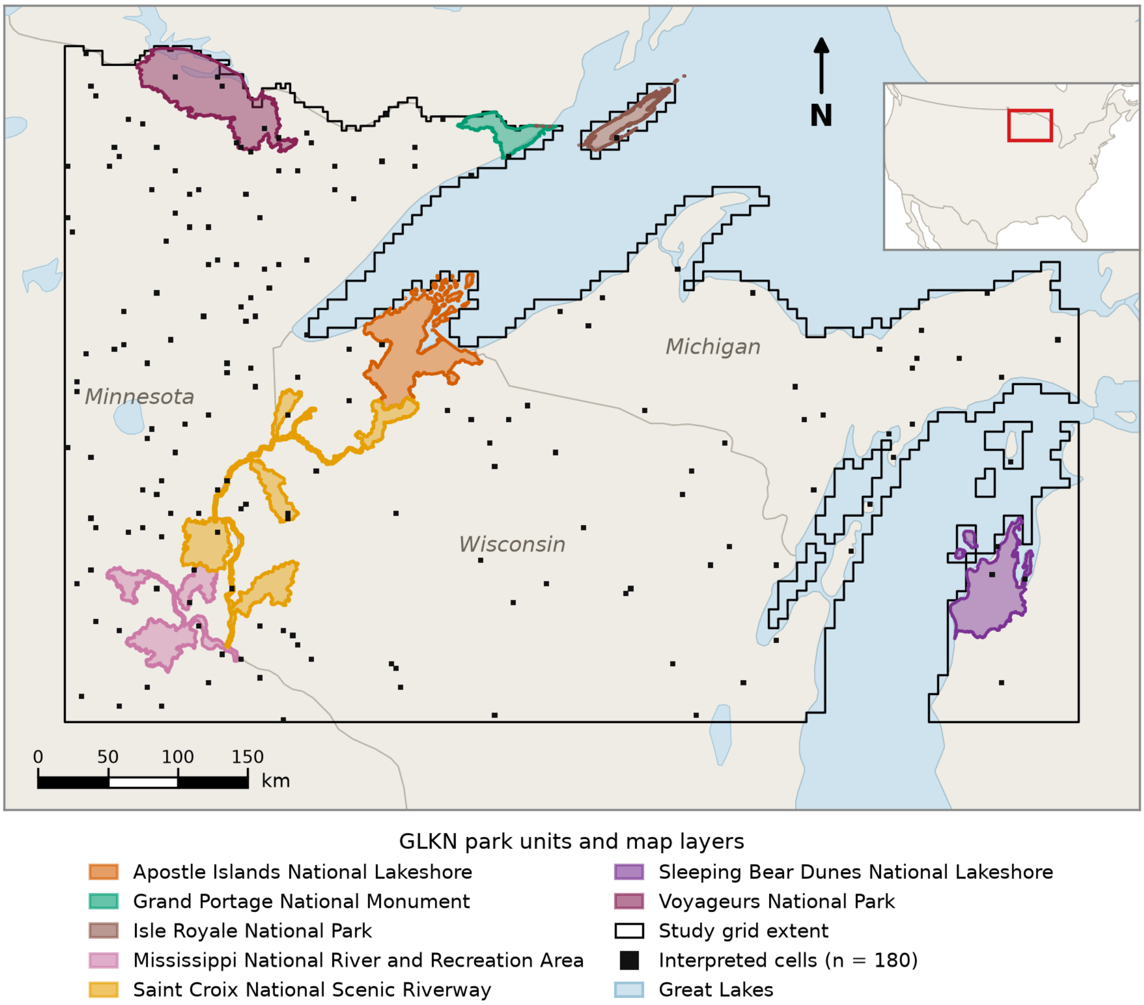
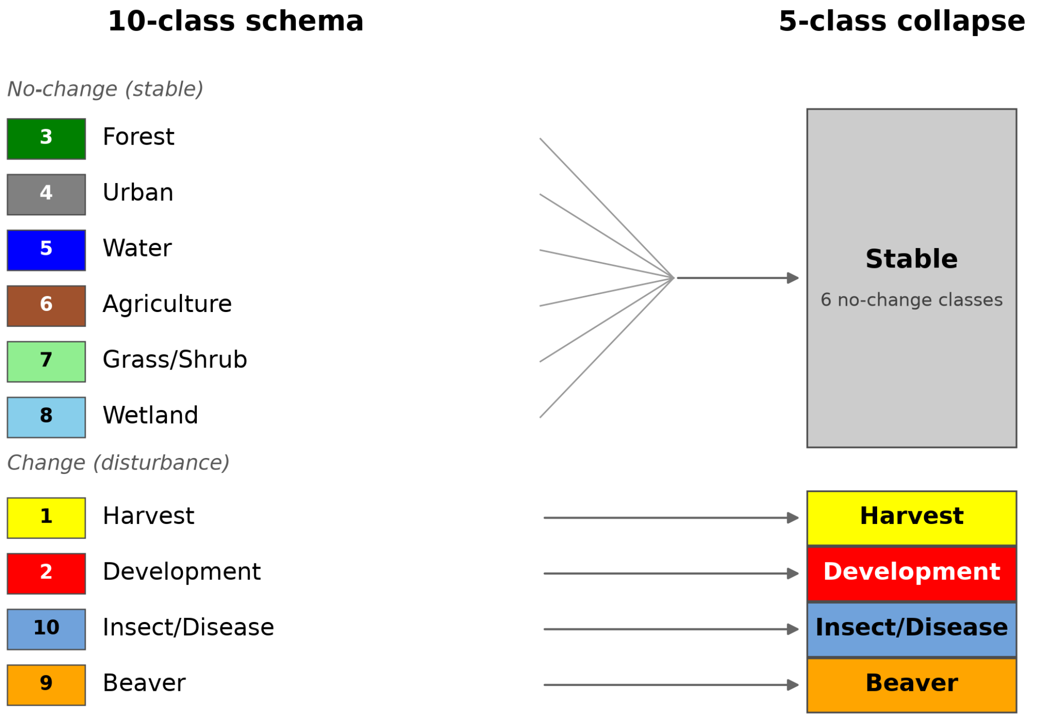


Figure 1. Study area in the western Great Lakes region (Wisconsin, Minnesota, and the Michigan Upper Peninsula), EPSG:5070, showing the study grid extent, the Great Lakes, the seven GLKN park units, and the 180 interpreted reference cells, with a conterminous-US locator inset, scale bar, and north arrow.

2.2 Class schema and the 5-class collapse

The classification schema has 10 classes: Harvest, Development, Forest, Urban, Water, Agriculture, Grass/Shrub, Wetland, Beaver, and Insect/Disease. The CKIT interpreter label identifiers are crosswalked to this schema (0 to Water, 1 to Agriculture, 2 to Grass/Shrub, 3 to Forest, 4 to Urban, 5 to Wetland, 20 to Harvest, 30 to Development, 50 to Insect/Disease, 62 to Beaver). Two reference values have no 10-class equivalent and are dropped: 10 (Unknown, an abstention) and 13 (Other, no-change). Fire (40) carries no pixels in this reference set.

For change-focused analysis the schema is collapsed to five classes: Stable (the six no-change classes pooled), Harvest, Development, Insect/Disease, and Beaver. The primary 5-class collapse (used for the pooled comparison, the per-cell analyses, and the interpreter agreement) excludes Unknown and Other. A second per-bracket 5-class product folds Other into Stable instead; the two collapses are kept distinct and are not interchangeable (see the consistency report). Figure 1b shows the 10-class schema and the 5-class collapse.



Class code shown in each swatch. The 5-class collapse folds the six stable classes into Stable and keeps the four change classes.

Figure 1b. Classification schema. Left, the 10-class schema grouped into the six no-change (stable) classes and the four change (disturbance) classes, each with its class code and legend color. Right, the 5-class collapse that folds the six stable classes into a single Stable class and keeps the four change classes.

2.3 Prediction sources: embedding variants and the spectral baseline

The embedding predictions are Random Forest classifications of AlphaEarth Foundations embeddings, run in Google Earth Engine. Five embedding feature configurations are compared, distinguished by how the base-year and paired-year embeddings and their change are combined (base year 2018, paired year 2020 in the training window, shifting per bracket in the temporally-matched runs):

- v2: base + delta (change from base to paired year).
- v3: base + paired year (both years stacked).
- v4: delta only.
- v5: base + dot product.
- v6: dot product only.

The spectral baseline (spec_all) is a Random Forest trained on 2018/2020 spectral composites drawn from Sentinel-2, Landsat 8, and Sentinel-1 SAR raw bands and indices (50 bands). All six sources are classified into the same 10-class schema on the same cell footprints, so the embedding-versus- spectral comparison is like-for-like on the shared cells.

2.4 Temporal transferability design

A single classifier is trained once on the 2018/2020 window and applied to five NAIP brackets: 2017-2019, 2018-2020, 2019-2021, 2020-2022, and 2021-2023. The 2018-2020 bracket is the in-sample control (its own year window). The five brackets use disjoint 36-cell sets with no cell shared across brackets, so a bracket-to-bracket difference confounds temporal transfer with the differing cell composition and landscape difficulty. The per-bracket results are therefore five independent assessments rather than a controlled transfer curve, and the pooled numbers, or a same-bracket comparison, are the like-for-like readings. The embedding predictions cover all 180 cells; spec_all covers 168, since 12 spec_all rasters are entirely nodata in the out-of-sample brackets.

2.5 Accuracy assessment

For each source, bracket, and pooling, a confusion matrix is built with the interpreted reference on rows and the prediction on columns, counting only pixels where both carry a valid class. From each matrix we report overall accuracy (OA), Cohen's kappa, macro-F1, and mean IoU, and per class the user's accuracy (UA, precision), producer's accuracy (PA, recall), F1, IoU, and reference support (pixel count). Because pixels within a cell are spatially autocorrelated, per-class metrics also carry a design-aware low-support flag that fires when a class appears in fewer than five cells or has under 100 reference pixels, since a large pixel count concentrated in one or two cells is still only one or two independent observations.

2.6 Inter-interpreter reference reliability

The 72 double-interpreted cells are used to quantify how reliable the reference itself is, per class. For each reviewer pair a pixel-for-pixel confusion matrix is built on the shared footprint, and per class the agreement F1 (the balanced probability that the two interpreters concur given one assigned the class) is computed, both for the 10-class schema and the 5-class collapse. Confidence intervals use a cluster (pair) bootstrap: the resampling unit is the pair, not the pixel, so within-cell autocorrelation is respected. Classes are assigned a reliability tier on F1 (High at or above 0.70, Moderate 0.50 to 0.70, Low below 0.50).

Two supporting diagnostics separate the kinds of disagreement. A spatial-tolerance analysis recomputes per-class agreement under relaxed neighborhood matching, above a heterogeneity null, to isolate boundary-misregistration disagreement from conceptual disagreement. A directional-asymmetry analysis pools the directed reviewer confusion per class and reports an over-assignment index with pair-bootstrap CIs, to test whether reviewers carry systematic class leanings.

2.7 Model accuracy against the reliability ceiling

To place model accuracy against reference reliability, each source's per-class F1 under the 5-class collapse is computed against the adjudicated reference, pooled over the source's usable cells, with a cluster (cell) bootstrap. These model F1 values are shown alongside the inter-interpreter agreement F1 for the same class (the reliability ceiling from Section 2.6), so the gap between a source and the human ceiling is visible per class.

2.8 Per-cell analyses

Beyond pooled metrics, each grid cell is treated as one sample. For every source and cell, the 5-class confusion is built and the cell's macro-F1 is the mean of the per-class F1 over the classes present in the reference or the prediction for that cell. A per-class version restricts to each change class and reports its per-cell F1 distribution. The six sources are compared on the common set of cells usable for all six (168 cells); each source is also summarized on its own full usable set, which differs in N and is not a head-to-head ranking.

2.9 Spatial-structure diagnostics

Spatial coherence is measured within the interpreted cell footprints, so the reference sets the scale. Per source we report the mean patch size and the area-weighted median patch size from 8-connected component labeling, and Moran's I (queen contiguity) of the class raster as a smoothness diagnostic (class codes are nominal, so this is read as spatial smoothness, not autocorrelation of a meaningful variable). A complementary neighbor-change metric measures the fraction of horizontally-adjacent, both-valid pixel pairs whose class differs, over the full rasters.

2.10 Training-cap sensitivity

To probe the sensitivity of the rare change classes to training size, a v2 classifier is trained with the four change classes capped at 50, 100, 150, and 200 training points (stable classes held at 200) and applied per bracket, pooled over the 180 cells. Per change class we report UA, PA, F1, predicted pixel count, reference support, and the total unique-training-pixel ceiling, since the cap constrains the small-pool classes (Beaver about 502 pixels, Insect/Disease about 662) far more than the large-pool classes (Harvest about 18,800, Development about 8,500).

2.11 Software and reproducibility

All analyses are scripted (`scripts/`) and their curated outputs, with rendered figures and per-figure captions, are in `reports/`. Confusion matrices, metric tables, and figures regenerate from the raw predictions and the adjudicated reference; the raw prediction and sample sidecars are fetched from cloud storage and are not committed.

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3. Results

3.1 Overall accuracy by prediction source

Pooled over the five brackets, overall accuracy separates the smooth context-preserving embeddings from the change-only and dot-product configurations (Table 1, Figure 2). Among the embeddings, v2 attains the highest pooled OA (0.659) and v6 the lowest (0.130); the spectral baseline spec_all sits at 0.588 on its 168-cell pool. Kappa, macro-F1, and mean IoU follow the same ordering. `[Interpretation: relate the v2/v3/v5 versus v4/v6 split to the context-preserving versus change-only feature design.]`

Source	OA	Kappa	Macro-F1	Mean IoU	N cells	N pixels
v2	0.659	0.536	0.383	0.301	180	20,404,977
v3	0.635	0.514	0.381	0.299	180	20,404,977
v4	0.245	0.123	0.162	0.096	180	20,404,977
v5	0.593	0.472	0.365	0.285	180	20,404,977
v6	0.13	0.04	0.089	0.049	180	20,404,977
spec_all	0.588	0.461	0.362	0.275	168	18,756,059

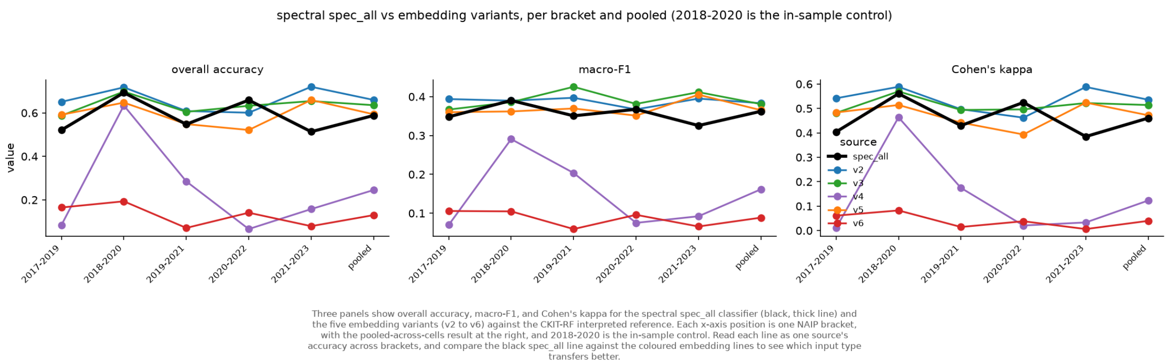
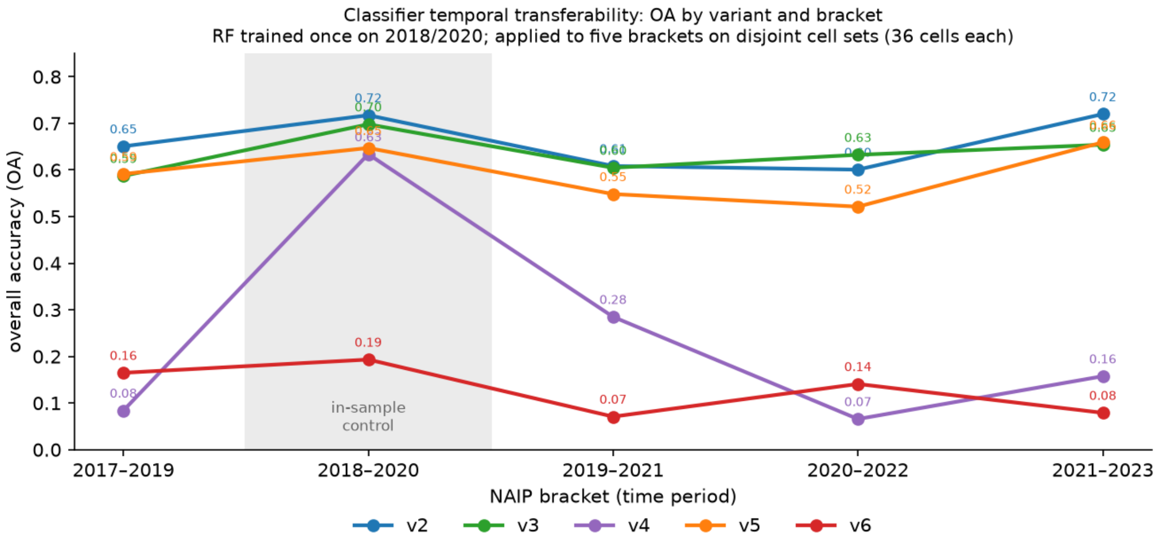


Figure 2. Overall accuracy, kappa, macro-F1, and mean IoU by prediction source (10-class schema, pooled; embeddings on 180 cells, spec_all on 168).

3.2 Accuracy across temporal brackets

Overall accuracy by source and bracket is shown in Table 2 and Figure 3, with the 2018-2020 in-sample control marked. The brackets use disjoint cell sets, so these are five independent assessments rather than a transfer curve. `[Interpretation: comment on which sources hold up off the training window and which do not, keeping the disjoint-cell caveat in view.]`

Source	2017-2019	2018-2020 (control)	2019-2021	2020-2022	2021-2023	Pooled
v2	0.651	0.717	0.608	0.601	0.72	0.659
v3	0.587	0.698	0.605	0.632	0.654	0.635
v4	0.084	0.634	0.285	0.066	0.158	0.245
v5	0.591	0.647	0.548	0.521	0.659	0.593
v6	0.165	0.193	0.071	0.141	0.079	0.13
spec_all	0.521	0.693	0.548	0.659	0.513	0.588



Overall accuracy of each RF variant (v2 to v6) when the single classifier trained on the 2018/2020 embeddings is applied to five NAIP brackets, with the time period on the x axis and OA on the y axis. Each colored line is one variant, dots carry the OA value, and the shaded 2018-2020 column marks the in-sample control. The five brackets use disjoint cell sets of 36 cells each, so read the points as five independent assessments, and not as a controlled transfer curve.

Figure 3. Overall accuracy per source across the five NAIP brackets (10-class schema, adjudicated reference; 2018-2020 is the in-sample control).

3.3 Per-class accuracy, 10-class schema

Per-class F1, pooled, is given in Table 3, with per-class UA, PA, F1, IoU, and support for every source in the supplementary long table (Table S1). Reference support is reported per class, and is low for the rare change classes, so their per-class numbers rest on little data. `[Interpretation: identify the classes that drive the aggregate differences.]`

Class	**v2**	**v3**	**v4**	**v5**	**v6**	**spec_all**	**Support (emb, 180)**	**Support (spec, 168)**
Harvest	0.119	0.083	0.146	0.041	0.023	0.102	232,769	201,530
Development	0.018	0.007	0.005	0.006	0.004	0.007	12,596	12,421
Forest	0.793	0.767	0.351	0.718	0.188	0.77	10,569,560	9,491,431
Urban	0.466	0.497	0.052	0.481	0.041	0.453	661,146	628,665
Water	0.938	0.938	0.3	0.933	0.15	0.94	1,725,908	1,490,487
Agriculture	0.8	0.806	0.477	0.797	0.265	0.62	3,110,297	3,086,878
Grass/Shrub	0.246	0.28	0.168	0.256	0.099	0.332	1,768,935	1,685,451
Wetland	0.431	0.409	0.097	0.405	0.109	0.373	2,247,007	2,086,285
Beaver	0.004	0.003	0.012	0.003	0.001	0.004	8,021	7,922
Insect/Disease	0.016	0.017	0.008	0.012	0.006	0.023	68,738	64,989

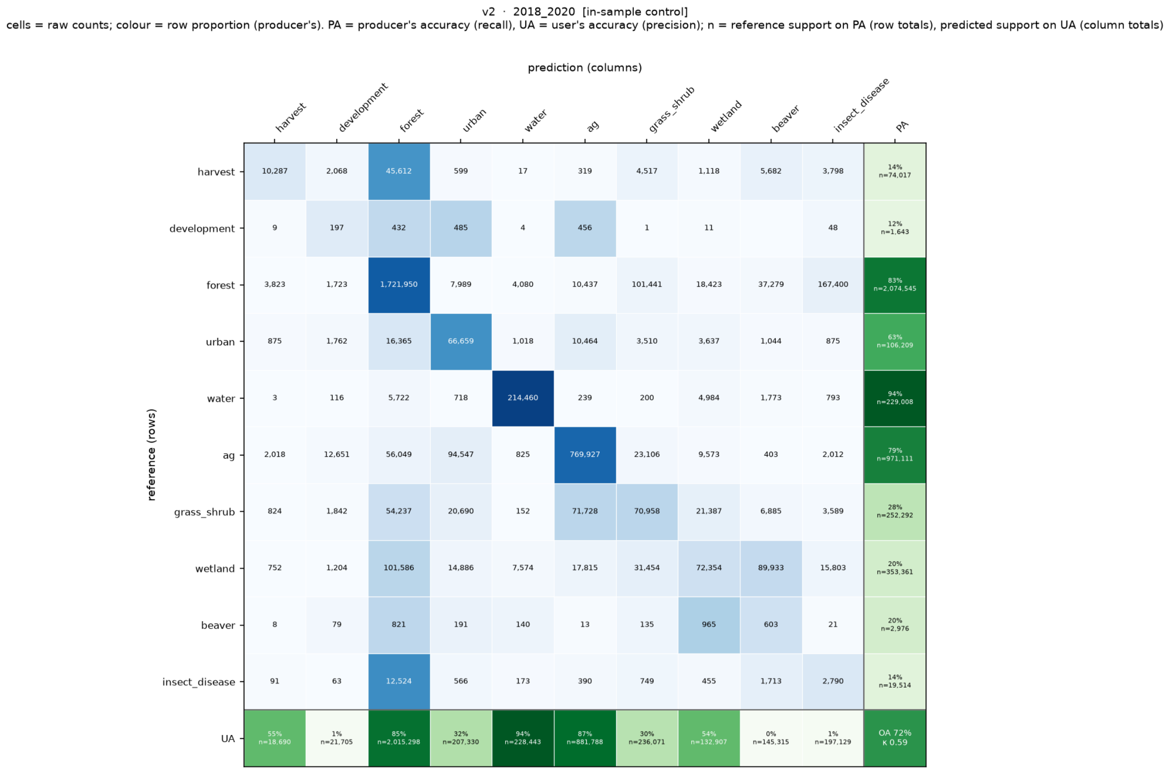


Figure 5a. Per-class confusion for v2, in-sample control bracket 2018-2020 (10-class, 36 cells; raw counts colored by row proportion, PA column and UA row, OA and kappa in the corner). See the consistency report on the per-bracket versus pooled basis.

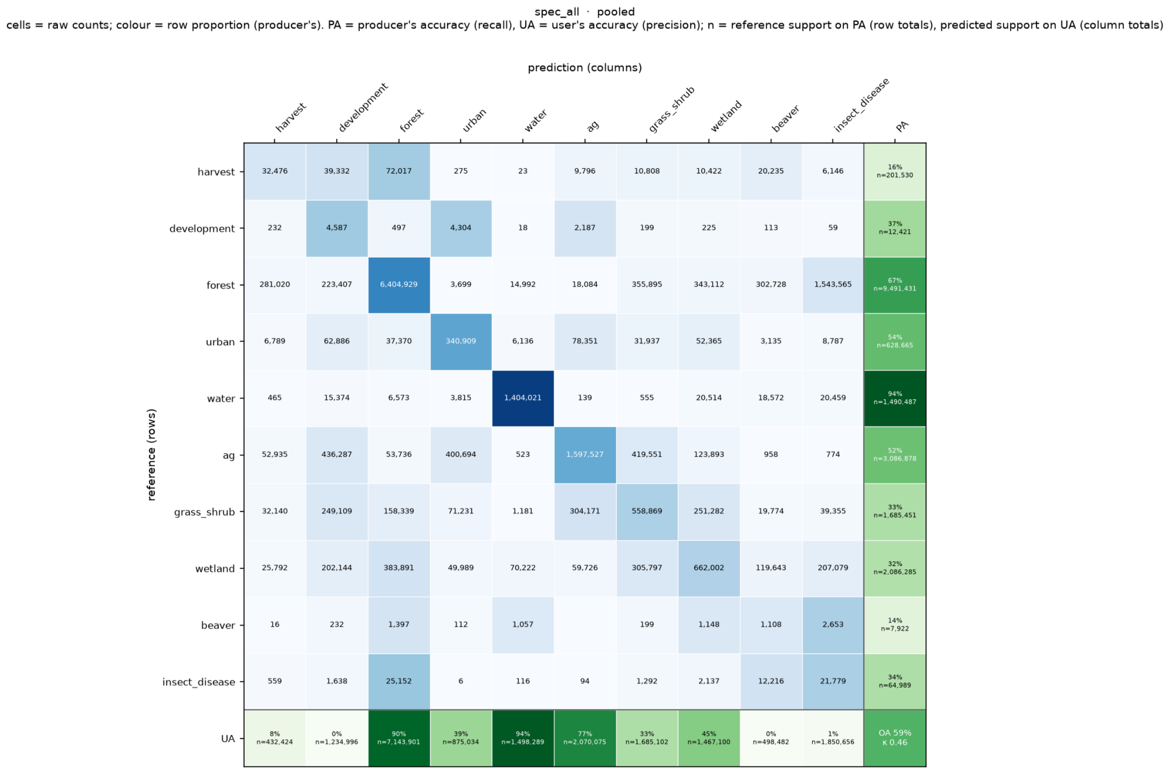


Figure 5b. Per-class confusion for spec_all, pooled (10-class, 168 cells).

3.4 Change-focused accuracy under the 5-class collapse

Under the 5-class collapse on the common 168-cell set, overall accuracy is high for every source because the landscape is stable-dominated (the all-Stable baseline OA is 0.985), so kappa and macro-F1 carry the change signal (Table 4, Figure 4). The pooled 5-class OA is highest for v4 (0.897) and lowest for v6 (0.602). `[Interpretation: contrast the 5-class OA ordering with the 10-class ordering in Table 1, noting the role of the stable-dominated baseline and of v4's behavior when the change classes are folded.]`

Source	OA	All-Stable baseline OA	Kappa	Macro-F1	Mean IoU	N cells
v2	0.875	0.985	0.039	0.22	0.193	168
v3	0.833	0.985	0.02	0.201	0.177	168
v4	0.897	0.985	0.071	0.223	0.198	168
v5	0.799	0.985	0.012	0.191	0.167	168
v6	0.602	0.985	0.004	0.157	0.124	168
spec_all	0.782	0.985	0.031	0.203	0.171	168

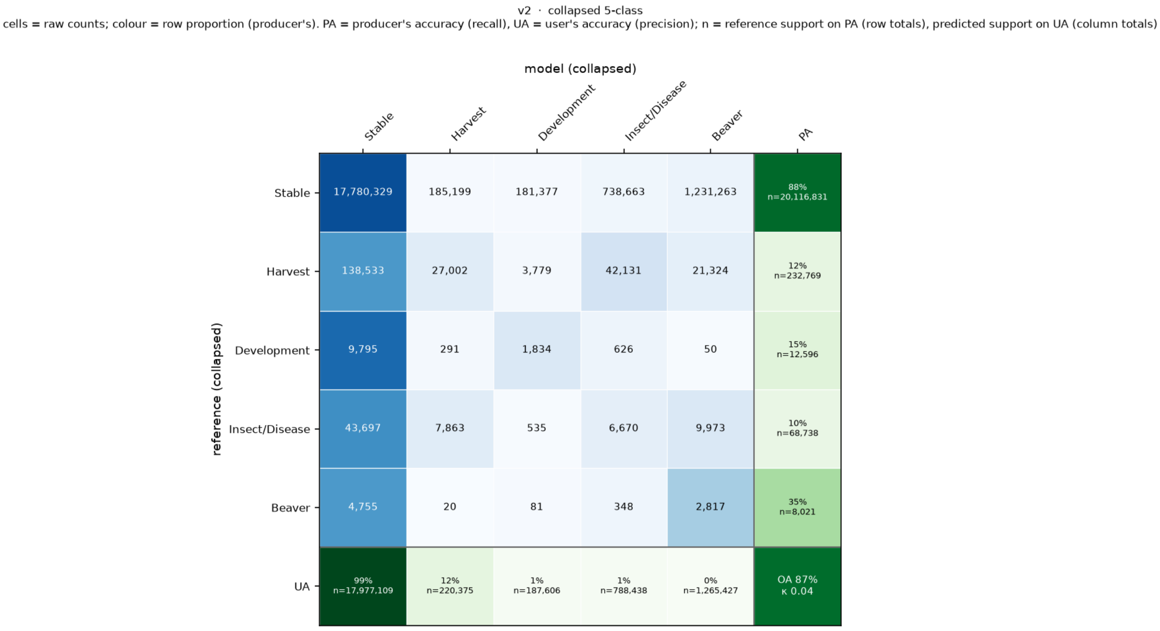


Figure 4a. Pooled 5-class confusion for v2 (180 cells).

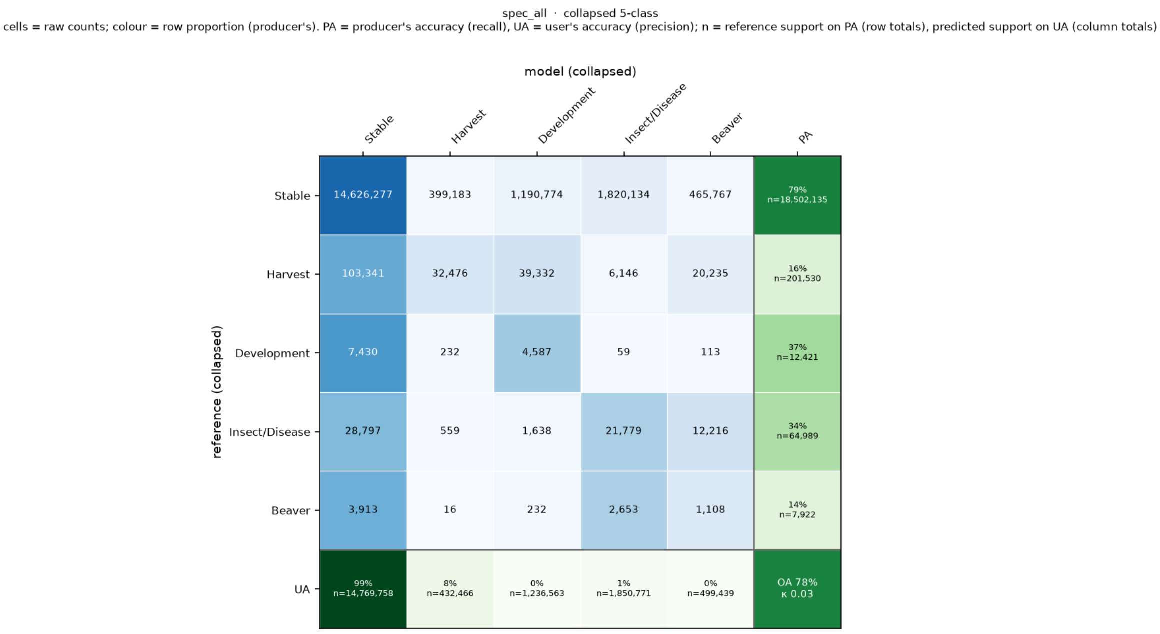
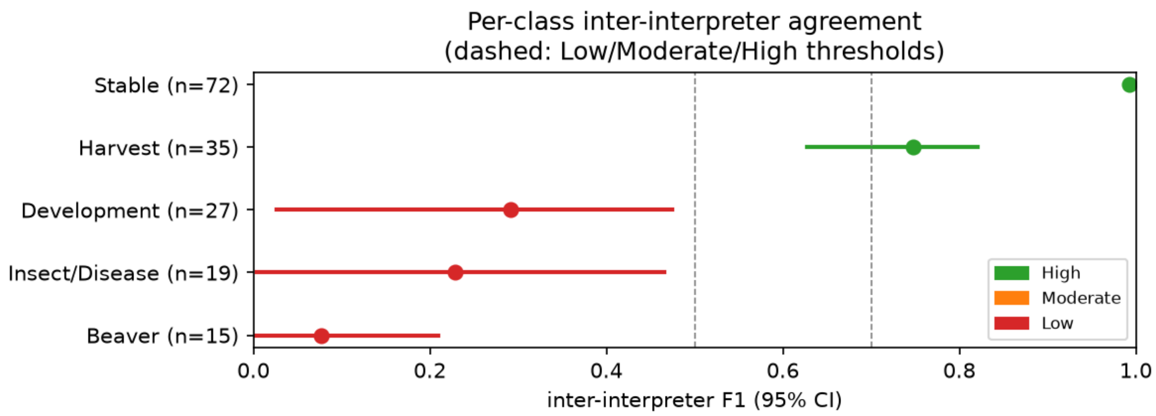


Figure 4b. Pooled 5-class confusion for spec_all (168 cells).

3.5 Reliability of the interpreted reference

Inter-interpreter agreement over the 72 double-interpreted cells sets a per-class reliability ceiling (Tables 5 and 6, Figures 6 and 7). Mean per-pair overall agreement is 0.77 (kappa 0.60). In the 5-class collapse, interpreters agree almost perfectly on Stable (F1 0.993) and well on Harvest (0.749), and fall to Low reliability on Development (0.295), Insect/Disease (0.229), and Beaver (0.077). In the 10-class schema, Water, Forest, and Agriculture are High reliability while Grass/Shrub and Wetland are Low. `[Interpretation: state the consequence for model evaluation on the Low-reliability classes.]`

Class	N Pairs	Support (px)	F1	F1 95% CI	IoU	IoU 95% CI	Reliability
Stable	72	8,087,379	0.993	0.99-0.996	0.986	0.981-0.991	High
Harvest	35	124,641	0.748	0.628-0.821	0.597	0.458-0.696	High
Development	27	9,288	0.292	0.026-0.475	0.171	0.013-0.312	Low
Insect/Disease	19	56,257	0.229	0.004-0.466	0.129	0.002-0.303	Low
Beaver	15	8,828	0.077	0.0-0.209	0.04	0.0-0.117	Low



Forest plot of per-class inter-interpreter agreement F1 for each land-cover class, where each dot is the pooled point estimate and its horizontal bar is the 95% cluster (pair) bootstrap confidence interval. F1 is the balanced probability that the two interpreters concur given one assigned the class, and the dashed vertical lines mark the Low, Moderate, and High reliability thresholds at 0.50 and 0.70. Dots colored orange or red identify classes such as Beaver and Insect/Disease where the human reference itself is unreliable.

Figure 7. Inter-interpreter per-class agreement F1 with 95% cluster (pair) bootstrap CIs, 5-class collapse, 72 pairs.

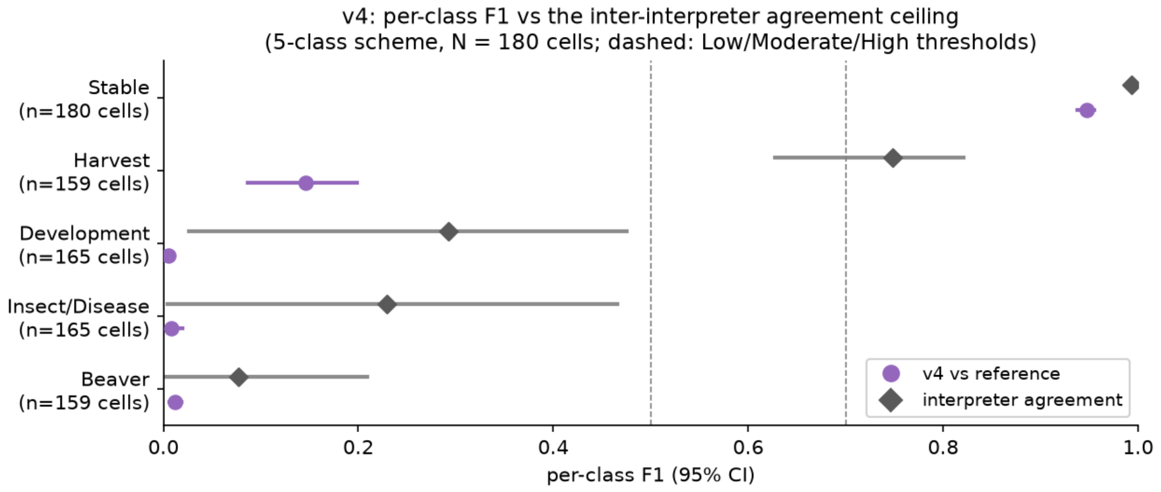
The full 10-class agreement table (Table 5) and its forest plot (Figure 6) are reported in the supplement-adjacent material; the pooled interpreter confusion matrix is Figure S1.

3.6 Model accuracy against the reliability ceiling

Placing each source's per-class 5-class F1 next to the inter-interpreter ceiling shows the two regimes per class (Table 7, Figure 8). On Stable the sources approach the ceiling (v4 0.947 versus 0.993). On Harvest the best source reaches 0.146 (v4) against a ceiling of 0.749, a large gap. On Development, Insect/Disease, and Beaver both the sources and the ceiling are low. `[Interpretation: distinguish the reducible gap on Harvest from the reference-limited classes.]`

Class	v2	v3	v4	v5	v6	spec_all	Interpreter ceiling
Stable	0.933	0.907	0.947	0.886	0.749	0.879	0.993
Harvest	0.119	0.083	0.146	0.041	0.023	0.102	0.748
Development	0.018	0.007	0.005	0.006	0.004	0.007	0.292
Insect/Disease	0.016	0.017	0.008	0.012	0.006	0.023	0.229

Beaver	0.004	0.003	0.012	0.003	0.001	0.004	0.077



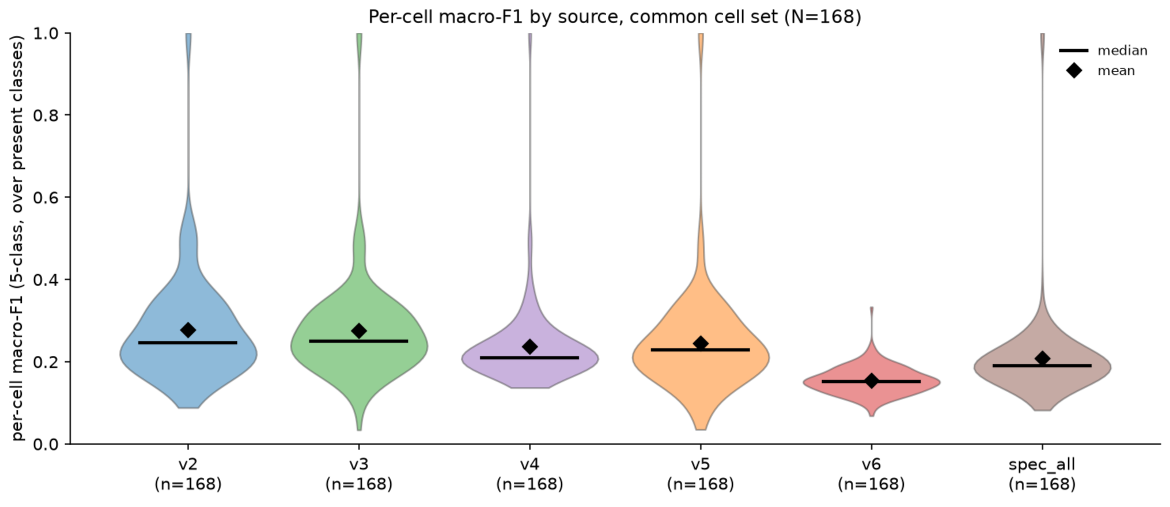
Per-class F1 in the 5-class collapsed scheme for v4 (colored circle) scored against the adjudicated interpreted reference, next to the inter-interpreter agreement for the same class (grey diamond), each with its 95% bootstrap confidence interval. The model uses a cluster (cell) bootstrap and the interpreter a cluster (pair) bootstrap. The interpreter bar is the reliability ceiling: where two humans barely agree, such as Development, Insect/Disease, and Beaver, the model score is bounded by reference noise rather than model error alone, so the gap to the grey diamond, not the absolute F1, is the reducible part.

Figure 8. Per-class F1 for v4 (colored) against the inter-interpreter ceiling (grey), 5-class collapse, with 95% bootstrap CIs. One panel per source in reports/model_vs_interpreter_5class/.

3.7 Per-cell distributions

The per-cell macro-F1 distributions (Table 10, Figure 11) and the per-class change-class F1 distributions (Table 11, Figure 12) summarize how each source does cell by cell rather than in the pool. `[Interpretation: note where a distribution spikes at zero for a change class and what that indicates about commission or omission.]`

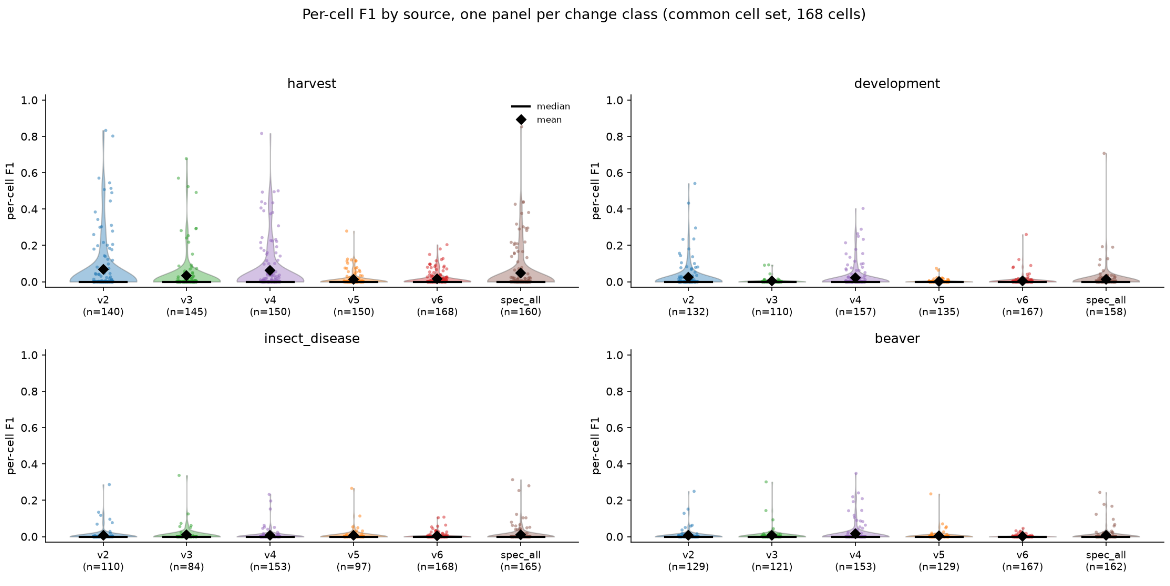
Source	N (common)	Mean F1 (common)	Median F1 (common)	N (full)	Mean F1 (full)	Median F1 (full)
v2	168	0.278	0.246	180	0.277	0.247
v3	168	0.275	0.25	180	0.273	0.249
v4	168	0.236	0.21	180	0.24	0.21
v5	168	0.244	0.228	180	0.24	0.225
v6	168	0.155	0.152	180	0.154	0.151
spec_all	168	0.207	0.19	168	0.207	0.19



Per-cell 5-class macro-F1 for each source on the common cell set, one point per cell, with the median (bar) and mean (diamond) marked. A class is included in a cell's macro-F1 if it is present in reference or prediction. All six sources are scored on the identical cells, so the distributions are directly comparable; the change-class commission drags the macro-F1 down whenever a source over-predicts a change class the cell does not contain.

Figure 11. Per-cell 5-class macro-F1 by source on the common 168-cell set, median (bar) and mean (diamond) marked.

Change class	v2	v3	v4	v5	v6	spec_all
Harvest	0.067	0.034	0.062	0.012	0.016	0.048
Development	0.025	0.003	0.022	0.002	0.006	0.013
Insect/Disease	0.008	0.009	0.006	0.006	0.005	0.011
Beaver	0.007	0.006	0.015	0.004	0.002	0.007



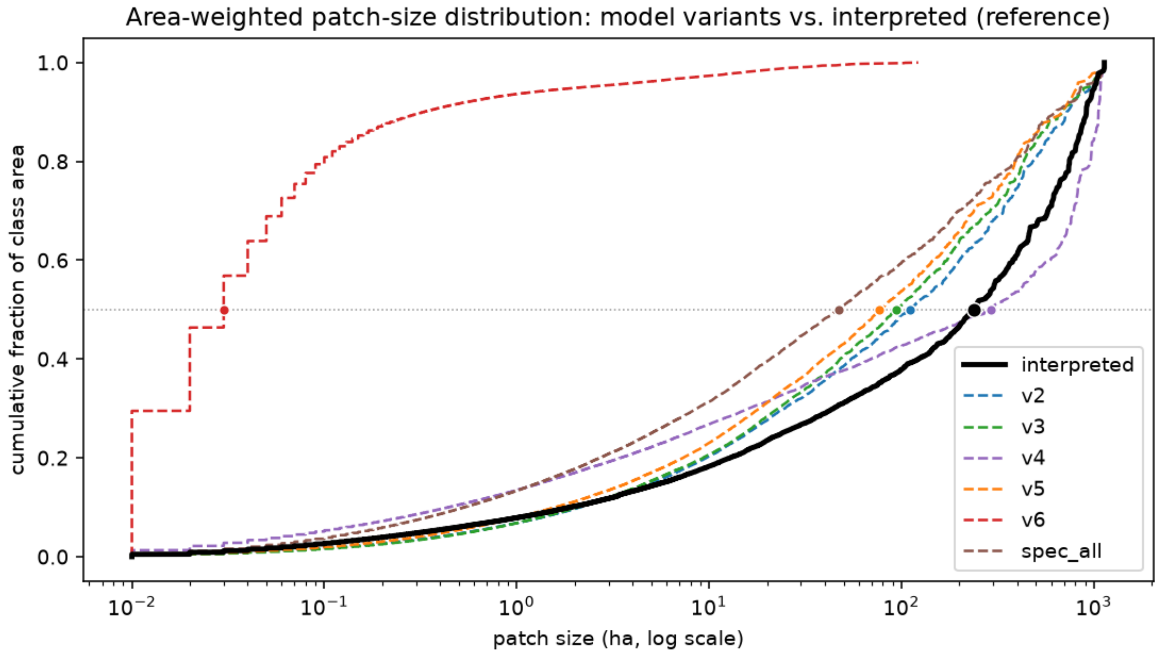
Distribution across cells of each change class's per-cell F1, one panel per class and six sources per panel, with individual cells as points, the median as a bar, and the mean as a diamond. A cell contributes to a class only where the class is present (reference or prediction), so the per-source N differs and is annotated under each source; a distribution over a handful of cells is not comparable in size to one over many. A spike at 0 means the class is mostly commission or omission there.

Figure 12. Per-cell F1 for the four change classes by source (5-class collapse, common set; contributing cell counts annotated).

3.8 Spatial structure and speckle

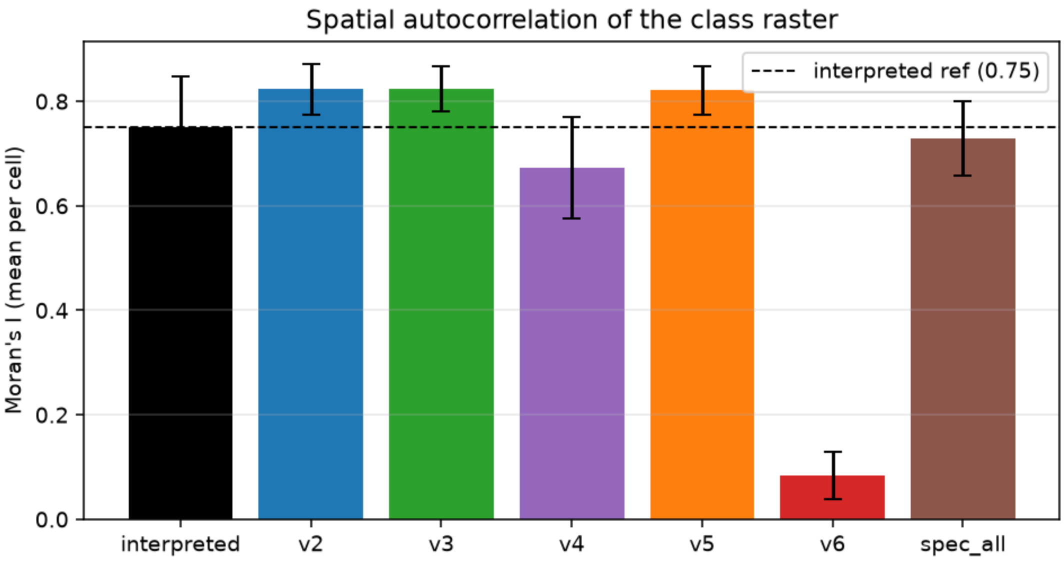
The spatial-structure diagnostics (Table 9, Figure 9) and the neighbor-change speckle metric (Table S3, Figure 10) characterize the maps' spatial coherence. The interpreted reference has a mean patch size of 0.79 ha and Moran's I 0.75; among the sources, v6 is the outlier (mean patch 0.02 ha, Moran's I 0.08, neighbor-change 0.781), consistent with its per-pixel dot-product format, while v2, v3, and v5 cluster near the reference scale. [Interpretation: connect the spatial-structure ordering to the accuracy ordering and to the point that similar aggregate accuracy can hide different spatial predictions.]

Source	N patches	Mean patch (ha)	Median-by-area (ha)	Moran's I (mean)	Moran's I (std)
interpreted (ref)	258,077	0.79	238.87	0.75	0.097
v2	186,504	1.1	110.38	0.82	0.048
v3	167,631	1.22	93.97	0.82	0.043
v4	553,745	0.37	290.65	0.67	0.098
v5	205,109	1	77	0.82	0.046
v6	9,502,626	0.02	0.03	0.08	0.045
spec_all	337,808	0.56	47.02	0.73	0.071



Cumulative fraction of class area held in patches at or below a given size (hectares, log scale), the same patches as the count ECDF but weighted by pixel area, so single-pixel specks collapse to the left and the grain of the landscape shows. The dot marks each source's median-by-area patch size, where half of its class area sits in smaller patches. Sources further right hold their area in larger, more contiguous patches; the interpreted reference (solid black) is the coarsest.

Figure 9a. Area-weighted patch-size ECDF by source (within interpreted footprints).

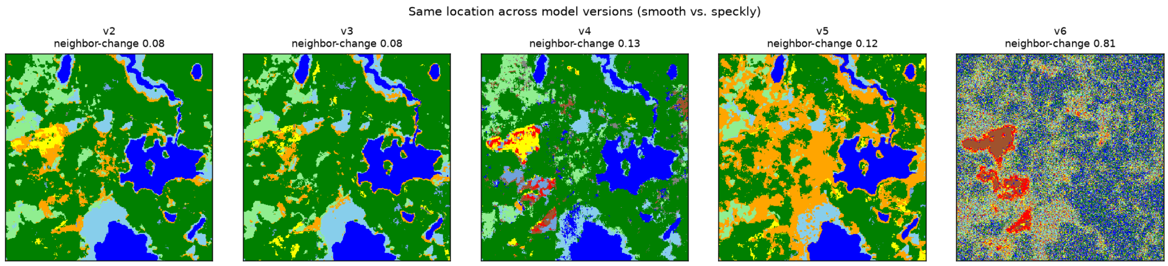


Mean per-cell Moran's I (queen contiguity over the class raster) for each source, with error bars for the between-cell standard deviation and the interpreted reference value marked (dashed). Higher Moran's I means more spatial autocorrelation (smoother maps); v6 is near zero (salt-and-pepper speckle), while the smoother classifiers sit close to the interpreted reference.

Moran's I of the class raster, by source

Spatial autocorrelation (Moran's I) of the class raster, per source.
Axes: x = source; y = mean per-cell Moran's I with queen (8-neighbor) contiguity; dashed line = interpreted reference. NB: class codes are nominal, so read this as a spatial-smoothness diagnostic, not autocorrelation of a meaningful quantitative variable.
Why: an alternative smoothness measure that complements patch size.
Interpret: v2/v3/v5 ~0.82 (smooth, near the interpreted 0.75), v4 lower, v6 ~0.09 (speckle). It cleanly isolates v6 but does not separate v2/v3/v5 — patch size does that.

Figure 9b. Moran's I by source.



Same location across model versions (smooth vs speckly)

The same map location shown across model versions.
Layout: one panel per version, identical crop, classes colored; each annotated with its local neighbor-change.
Why: visualize smooth vs speckly output.
Interpret: v2/v3/v5 are coherent patches, v4 grainier, v6 salt-and-pepper (the water body is barely discernible).

Figure 10. Cropped classified-map detail per embedding variant illustrating the neighbor-change speckle metric.

3.9 Training-cap sensitivity for the change classes

Table 8 and Figure 13 report how the change-class metrics move as the training cap varies from 50 to 200 points, with the training ceiling per class given so the cap is read relative to the available pool. `[Interpretation: state, for Beaver and Insect/Disease especially, whether lower caps trade commission for recall or simply relabel, per the interpretive guardrail in the analysis note.]`

Change class	Training cap	Training ceiling (px)	UA (precision)	PA (recall)	F1	Support (px)	Predicted (px)
Beaver	50	502	0.005	0.236	0.009	8,021	417,661
Beaver	100	502	0.003	0.301	0.005	8,021	963,501
Beaver	150	502	0.002	0.362	0.004	8,021	1,372,227

Beaver	200	502	0.002	0.351	0.004	8,021	1,264,957
Development	50	8,482	0.005	0.057	0.01	12,596	131,906
Development	100	8,482	0.006	0.171	0.012	12,596	340,212
Development	150	8,482	0.005	0.214	0.009	12,596	584,544
Development	200	8,482	0.01	0.146	0.018	12,596	187,381
Harvest	50	18,807	0.487	0.501	0.494	232,769	239,688
Harvest	100	18,807	0.292	0.646	0.402	232,769	515,011
Harvest	150	18,807	0.249	0.67	0.364	232,769	625,679
Harvest	200	18,807	0.123	0.116	0.119	232,769	220,320
Insect/Disease	50	662	0.008	0.08	0.015	68,738	672,730
Insect/Disease	100	662	0.008	0.083	0.014	68,738	720,331
Insect/Disease	150	662	0.007	0.085	0.013	68,738	795,788
Insect/Disease	200	662	0.008	0.097	0.016	68,738	787,829

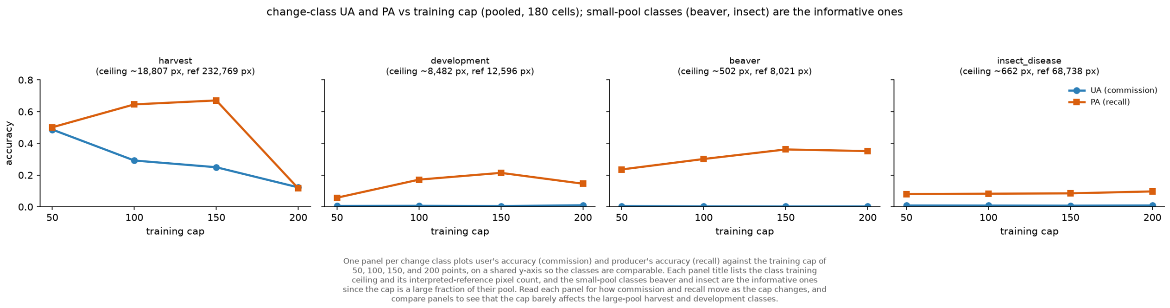


Figure 13. User's and producer's accuracy for the four change classes versus the training cap (v2, 10-class, 180 cells).

3.10 Robustness

The choice of which reviewer represents a multi-interpreted cell was tested by repeating the pick-one-per-location draw 100 times on the earlier snapshot; the version ordering is stable (Table S2). Map speckle by variant is reported in Table S3.

version	metric	mean	std	min	max	p2_5	p97_5	range
v2	overall_accuracy	0.657	0.0019	0.6517	0.6614	0.6535	0.6606	0.0097
v2	macro_f1	0.3793	0.0018	0.3752	0.3835	0.3758	0.3826	0.0083
v2	mean_iou	0.2979	0.0016	0.2943	0.302	0.2952	0.3011	0.0077
v2	kappa	0.5303	0.0026	0.5232	0.5363	0.5255	0.535	0.0131
v3	overall_accuracy	0.6062	0.0017	0.6017	0.6101	0.6029	0.6093	0.0084
v3	macro_f1	0.3724	0.0018	0.3685	0.3764	0.3693	0.3759	0.0078
v3	mean_iou	0.2907	0.0016	0.2873	0.2944	0.2882	0.2939	0.0071
v3	kappa	0.4833	0.0023	0.4776	0.4885	0.4793	0.4873	0.0109
v4	overall_accuracy	0.5071	0.0018	0.5025	0.5117	0.5033	0.5104	0.0091
v4	macro_f1	0.2431	0.0016	0.2398	0.2468	0.2402	0.2458	0.007
v4	mean_iou	0.1632	0.0011	0.161	0.1658	0.1612	0.1651	0.0048
v4	kappa	0.3186	0.0023	0.3132	0.3237	0.314	0.3225	0.0105
v5	overall_accuracy	0.5643	0.0017	0.5602	0.5683	0.5611	0.5671	0.0081
v5	macro_f1	0.361	0.0017	0.357	0.3648	0.358	0.3643	0.0077
v5	mean_iou	0.2793	0.0015	0.2758	0.283	0.2769	0.2823	0.0072
v5	kappa	0.4418	0.0023	0.4367	0.447	0.4379	0.4454	0.0103

v6	overall_accuracy	0.1864	0.0004	0.1853	0.1874	0.1856	0.1873	0.0021
v6	macro_f1	0.1036	0.0004	0.1027	0.1044	0.1028	0.1043	0.0017
v6	mean_iou	0.0582	0.0002	0.0577	0.0587	0.0578	0.0586	0.001
v6	kappa	0.0591	0.0005	0.0574	0.0604	0.0582	0.06	0.0029

End of draft. Sections marked `[Interpretation:]` and the missing study-area figure (Figure 1) are left for the author. See `manuscript_tables_figures_plan.md` for the full candidate set and `consistency_report.md` for the basis caveats that any combined statement must respect.